SMART WATER USAGE MONITOR

Multi-Step Prompt Workflow for Practical Household Water Conservation

NAME : D.Martina Angelin

EMAIL : 99230041121@klu.ac.in

REGNO : 99230041121

01 | Concept

Purpose: Smart Water Usage Monitor helps households understand their daily water-use activities and identify simple opportunities to reduce unnecessary consumption.

Problem: People may use water inefficiently during bathing, washing, cooking and cleaning without realizing where wastage occurs. A structured AI workflow can convert daily usage information into practical conservation suggestions.

02 | Objective

The workflow uses sequential prompts to understand household water-use activities, identify high-usage areas, suggest conservation actions, create a practical daily plan, and review progress. Each stage uses relevant information produced by earlier stages.

03 | Workflow Overview

INPUT 1	**Understand**	**2**	**Identify**	**3**	**Suggest**	**4**	**Plan**	**5**	**Review**	**OUTPUT**

04 | Workflow Steps

1. Household Water-Use Analysis

Build a simple profile from daily activities, frequency, duration and estimated usage. Output: Usage profile + attention areas.

2. High-Usage Identification

Rank activities using frequency, duration and possible wastage. Output: Priority usage list.

3. Conservation Suggestions

Recommend simple actions for the highest-priority activities, such as reducing unnecessary flow and avoiding avoidable wastage. Output: Action list.

4. Daily Water Plan

Create a realistic routine for applying the recommended actions across daily household activities. Output: Day-by-day conservation plan.

5. Progress Review

Compare the household's progress with the plan, identify successful actions and remaining problem areas, and adjust the recommendations. Output: Updated plan.

05 | Prompt Workflow

Step 1 — Understand Household Water Use

Analyze the household's daily water-use activities, frequency, duration and estimated usage. Organize the information into a concise usage profile. Identify missing information without inventing details.

Step 2 — Identify High-Usage Activities

Using the usage profile, classify activities by frequency, duration and possible wastage. Rank the activities from highest to lowest priority and briefly explain the reason for each priority.

Step 3 — Generate Conservation Suggestions

Create practical water-saving actions for the prioritized activities. Keep the suggestions simple, realistic and suitable for normal household routines.

Step 4 — Build the Daily Water Plan

Create a realistic daily plan using the recommended actions. Organize the actions by household activity and avoid an unrealistic workload. Present the plan clearly.

Step 5 — Review and Adapt

Review progress against the water-saving plan. Identify completed actions, continuing wastage and upcoming priorities. Adjust the remaining recommendations while keeping them practical.

06 | Prompt Chaining

Step 1 produces the water-use profile; Step 2 uses it to identify high-usage activities; Step 3 uses those priorities to create conservation actions; Step 4 uses the actions to create a daily plan; and Step 5 uses progress information to revise the remaining plan.

STAGE	INPUT / CONTEXT	OUTPUT
1	Daily activities + usage details	Water-use profile
2	Step 1 profile	Priority usage list
3	Step 2 priorities	Conservation actions
4	Step 3 actions	Daily water plan
5	Progress + earlier context	Updated plan

07 | Demonstration Input

Example: A household has four members and uses water for bathing, washing clothes, cooking, cleaning and gardening. The family notices higher usage during long showers and outdoor cleaning and wants a simple daily conservation plan.

08 | Expected Final Output

A household water-use profile, prioritized high-usage activities, practical conservation actions, daily water-saving targets, progress indicators and recommendations for adjusting the plan. The result should be practical, balanced and easy to follow.

09 | Workflow Result & Evaluation

The completed workflow transforms unstructured household water-use information into a structured conservation strategy. Its final result is a connected planning cycle that can respond to changing household habits and progress.

INPUT	AI PROCESS	FINAL RESULT
Activities • Frequency • Duration • Usage	Analyze → Identify → Suggest → Plan → Review	• Priority activities • Conservation actions • Daily plan • Progress adjustments

10 | Sample Final Water-Saving Plan

DAY	FOCUS	ACTIVITY	OUTPUT
Day 1	Bathing	Reduce unnecessary water flow + track duration	Usage check
Day 2	Laundry	Use planned washing cycles + avoid excess water	Laundry target
Day 3	Cleaning	Use water carefully during floor/outdoor cleaning	Wastage check
Day 4	Gardening	Water plants at an appropriate time and amount	Garden target
Day 5	Review	Check all activities + weak areas	Updated plan

11 | Key Benefits

• Personalized planning based on household water-use habits

• Priority-driven identification of high-usage activities

• Integrated conservation actions and daily planning

• Progress-aware adjustment of recommendations

• Clear, reusable prompt structure

• No coding required for the workflow design

12 | Conclusion

Smart Water Usage Monitor demonstrates how prompt engineering can break a practical household water-conservation problem into connected, measurable stages. By combining analysis, prioritization, conservation suggestions, planning and feedback, the workflow produces a practical strategy that can be refined as household habits change.